1. **In K-fold cross-validation, how will increasing *k* affect the variance (across subsamples) of estimated model parameters?**

- ◯ Increasing *k* will not affect the variance of estimated parameters.
- ◯ Increasing *k* will usually increase the variance of estimated parameters.
- ◯ Increasing *k* will usually reduce the variance of estimated parameters.
- ◯ Increasing *k* will increase the variance of estimated parameters if models are underfit, but reduce it if models are overfit.

2. **Which statement about K-fold cross-validation below is TRUE?**

- ◯ Each subsample in K-fold cross-validation has at least *k* observations.
- ◯ Each of the *k* subsamples in K-fold cross-validation is used as a training set.
- ◯ Each of the *k* subsamples in K-fold cross-validation is used as a test set.
- ◯ None of the above

3. **If a low-complexity model is underfitting during estimation, which of the following is MOST LIKELY true (holding the model constant) about K-fold cross-validation?**

- ◯ Assuming there is enough training data the number of folds in K-fold validation does not inherently result in underfitting.
- ◯ K-cross-validation with a small *k* will reduce or eliminate underfitting.
- ◯ K-fold cross-validation with a large *k* will reduce or eliminate underfitting.
- ◯ None of the above.

4. **Which of the following statements about a high-complexity model in a linear regression setting is TRUE?**

- ◯ Cross-validation with a small *k* will reduce or eliminate overfitting.
- ◯ A high variance of parameter estimates across cross-validation subsamples indicates likely overfitting.
- ◯ A low variance of parameter estimates across cross-validation subsamples indicates likely overfitting.
- ◯ Cross-validation with a large *k* will reduce or eliminate overfitting.